Eric Rosen

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Education

2018 – 2023 **PhD in Computer Science**, *Brown University*, Providence, RI
Research areas are in Artificial Intelligence, Machine Learning, and Robotics. Coursework included: Learning and Sequential Decision Making, Computer Vision, Computational Semantics, Computational Linguistics, Design and Implementation of Programming Languages

2018 – 2020 **Masters of Computer Science**, *Brown University*, Providence, RI

2014 – 2018 **Bachelor of Science in Computer Science and Applied Mathematics**, *Brown University*, Providence, RI
Received Bachelor's degrees with honors in Computer Science and Applied Mathematics. Coursework included: Applied Artificial Intelligence, Machine Learning, Neural Modeling Laboratory, Computing and Probability, Design and Analysis of Algorithms, Operations Research: Probabilistic Models, Operations Research: Deterministic Models, Digital Signal Processing

Work Experience

2023-current **Boston Dynamics AI Institute / RAI Research Scientist**, Boston, MA
Researched and developed algorithms for learning policies to perform bimanual object manipulation. Sources of data include human videos, teleoperated demonstrations, and online interaction with environment. Methods developed include behavior cloning and reinforcement learning techniques for data-driven generative models, and task and motion planning methods with structured 3D representations.

2022-2023 **Mitsubishi Electric Research Laboratories (MERL) Part-time Researcher**, Boston, MA
Researched algorithms for motor skill learning and Virtual Reality interfaces for human-robot interaction.

Fall 2021 **Facebook Reality Labs Part-time Researcher**, Redmond, WA
Worked on developing research articles and writing papers on deep learning for activity recognition.

Selected Publications

Journal Papers

IJRR 2019 **Communicating and Controlling Robot Arm Motion Intent Through Mixed-Reality Head-Mounted Displays**, *Eric Rosen, David Whitney, Daniel Ullman, Elizabeth Phillips, Stefanie Tellex*

Conference Papers

ICRA 2026 **Factorizing Diffusion Policies for Observation Modality Prioritization**, *Omkar Patil, Prabin Rath, Kartikay Pangaonkar, Eric Rosen, Nakul Gopalan*

ICRA 2025 **Verifiably following complex robot instructions with foundation models**, *Eric Rosen, Benedict Quartey, Stefanie Tellex, George Konidaris*

ICRA 2024 **Cape: Corrective actions from precondition errors using large language models**, *Shreyas Sundara Raman, Vanya Cohen, Ifrah Idrees, Eric Rosen, Raymond Mooney, Stefanie Tellex, David Paulius*

CORL 2023 **Synthesizing navigation abstractions for planning with portable manipulation skills**, *Eric Rosen, Steven James, Sergio Orozco, Vedant Gupta, Max Merlin, Stefanie Tellex, George Konidaris*

HRI 2022 **Learning reward functions from a combination of demonstration and evaluative feedback**, *Eric Hsiung, Eric Rosen, Vivienne Bihe Chi, Bertram F Malle*

IROS 2021 **Bootstrapping Motor Skill Learning with Motion Planning**, *Eric Rosen, Ben Abbatematteo, Stefanie Tellex, George Konidaris*

Best Paper Finalist IROS 2020 **Mixed Reality as a Bidirectional Communication Interface for Human-Robot Interaction**, *Eric Rosen, David Whitney, Michael Fishman, Daniel Ullman, Stefanie Tellex*

RSS 2020 **Simultaneously Learning Transferable Symbols and Language Groundings from Perceptual Data for Instruction Following**, *Nakul Gopalan, Eric Rosen, Stefanie Tellex, George Konidaris*

ICRA 2019 **End-User Robot Programming Using Mixed Reality**, *Samir Yitzhak Gadre, Eric Rosen, Gary Chien, Elizabeth Phillips, Stefanie Tellex, George Konidaris*

IROS 2018 **ROS Reality: A Virtual Reality Framework Using Consumer-Grade Hardware for ROS-Enabled Robots**, *Eric Rosen, David Whitney, Daniel Ullman, Elizabeth Phillips, Stefanie Tellex*

ICRA 2017 **Reducing Errors In Object-Fetching Interactions Through Social Feedback**, *David Whitney, Eric Rosen, James MacGlashan, Lawson LS Wong, Stefanie Tellex*

Magazine Articles

IEEE 2022 **A Tool for Organizing Key Characteristics of Virtual, Augmented, and Mixed Reality for Human–Robot Interaction Systems: Synthesizing VAM-HRI Trends and Takeaways**, *Thomas Groechel, Michael Walker, Christine T Chang, Eric Rosen, Jessica Forde*

Selected Awards and Honours

2019 **Hyundai Visionary Challenge Winner**, *Hyundai, Brown University*
A competition to accelerate research innovations in smart mobility that drive the creation of sustainable cities across the globe. Developed a multi-modal model for human-robot interaction involving goal specification and planning verification.

2019 **The IBM Watson AI XPRIZE Runner-up**, *XPRIZE Foundation*
A $5 million dollar challenge where 59 selected teams demonstrate how humans can collaborate with powerful artificial intelligence (AI) technologies to tackle some of the world's greatest challenges. Nominated and proposed a three-phase interdisciplinary research program to identify human social and moral norms and implement them in robots.

2019 **Analog Devices Realtime Sensor Fusion Challenge Runner-up**, *Analog Devices*
An innovation challenge for engineering students and startups to enhance context awareness in robotic system by fusing sensor data. Nominated and proposed solutions for fusing RGB-D, IMU, and thermal sensor data for enhancing the production of action maps in robot applications.

Internships

Summer 2021 **Facebook Reality Labs R&D Internship**, Redmond, WA
Developed deep learning design patterns and models for activity recognition using motion sensor for mixed reality platforms.

Summer 2020 **Uber ATG R&D Internship**, San Francisco, CA
Developed deep learning architectures to improve end-to-end motion planning for self-driving vehicles.

Summer 2017 **Amazon Robotics R&D Internship**, North Reading, MA
Researched machine learning techniques to improve robot manipulation in Amazon warehouses with multimodal feedback.

2014 – 2018 **Undergraduate Research Assistant**, *Brown University*, Providence, RI
Assisted and conducted published research in Humans to Robots Laboratory and Intelligent Robot Laboratory. ROS for programming robots, Java for probabilistic planning, Python for deep learning.

Service

ICRA24-26 **Associated Editor @ ICRA**, Handled papers related to AI-Enabled Robotics and Deep Learning in Grasping and Manipulation.

CoRL 2023 **Co-organizer @ Learning Effective Abstractions for Planning (LEAP) Workshop**

HRI20-23 **Co-organizer @ VAM-HRI Workshop**

Selected Teaching Experience

Spring 2022 **Instructor for Choreorobotics 101**, *Brown University*, Providence, RI
Created syllabus, course content, and curriculum for a new interdisciplinary course at Brown. Taught lectures on robotics and how to choreograph dances on the Boston Dynamic's Spot robot. Bridged vocabulary and concepts from choreography to robotics and helped students learn how to practically work with robots to create dance performances.